

Phyigital Samples: Enhancing Retail Experiences with Digital Augmentations of Physical Materials

Vincent van Rheden
Human-Computer Interaction
Division, University of Salzburg
Salzburg, Austria
vincent.vanrheden@plus.ac.at

Bernhard Maurer
Salzburg University of Applied
Sciences
Salzburg, Austria
bernhard.maurer@fh-salzburg.ac.at

Martin Murer
Human-Computer Interaction
Division, University of Salzburg
Salzburg, Austria
martin.murer@plus.ac.at

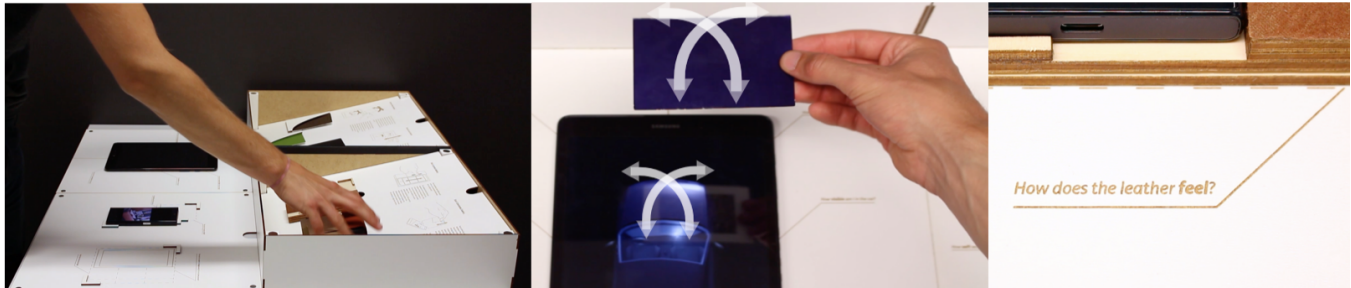


Figure 1: Phyigital sample box demonstrator; left: an overview of the demonstrator, middle: the phyigital interaction with a paint sample, right: provoking questions to focus on the detailed sample and product experience

ABSTRACT

Retail industry is subject to radical developments. Due to mass-customization it is not possible for retailers to present all possible configurations in-store for customer to experience. Additionally, there is a shift towards online shopping. Even in the context of purchasing cars it is common to pre-order cars online, without having experienced the car upfront. Physical material samples are a way for customers to get an impression of what the whole product is like, without retailers actually having the whole in their store. This comes with the drawback that it is hard for customers to experience the whole based on a single sample for a specific configuration. Particularly in the case of samples that have been taken home, the connection with the whole product can be lost. In this paper we present Phyigital Samples, combining physical and digital in the context of retail. Phyigital Samples connect with a digital layer that fills the gap between the detailed sample and the whole product. The interaction mapping with the sample and digital 'whole' or 'bigger picture' is based on the interaction to perceive the material quality of the specific sample. In this demo, we present the Phyigital Sample box containing four different samples illustrating our material interaction design approach.

CCS CONCEPTS

• **Human-centered computing** → **Interaction techniques.**

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
CHI EA '23, April 23–28, 2023, Hamburg, Germany
© 2023 Copyright held by the owner/author(s).
ACM ISBN 978-1-4503-9422-2/23/04.
<https://doi.org/10.1145/3544549.3583904>

KEYWORDS

Interaction Design, Materiality, Retail, Material Sample, Phyigital

ACM Reference Format:

Vincent van Rheden, Bernhard Maurer, and Martin Murer. 2023. Phyigital Samples: Enhancing Retail Experiences with Digital Augmentations of Physical Materials. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23)*, April 23–28, 2023, Hamburg, Germany. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3544549.3583904>

1 INTRODUCTION

The retail industry is developing radically in terms of products, customers, and sales channels. Contemporary industrial production allows for mass customization of configurations, materials, and finishes [12]. Consequently, from a retail perspective, the product variety makes it virtually impossible to carry the full catalog in a physical store. Sales channels change as well as e-commerce has grown alongside the internet, and accordingly, customers are increasingly informed about products they might purchase. However, customers might find it challenging to envision their desired product, in particular the experiential aesthetic qualities (haptics, visual, sound, etc) that one would get in reality visiting a physical store. Material samples aid in those situations, yet how might this be linked to the digital information customers have available when configuring a product? We present Phyigital Samples, a novel approach of connecting material samples with the digital realm. The aim is to create meaningful bridges between the physical sample and the product experience. In designing the interactions we took the material samples as a starting point. More specifically, we explored *how* one interacts with the sample in order to perceive the material qualities that the sample carries. For example, a velours leather sample carries tactile qualities. As explained by Lederman

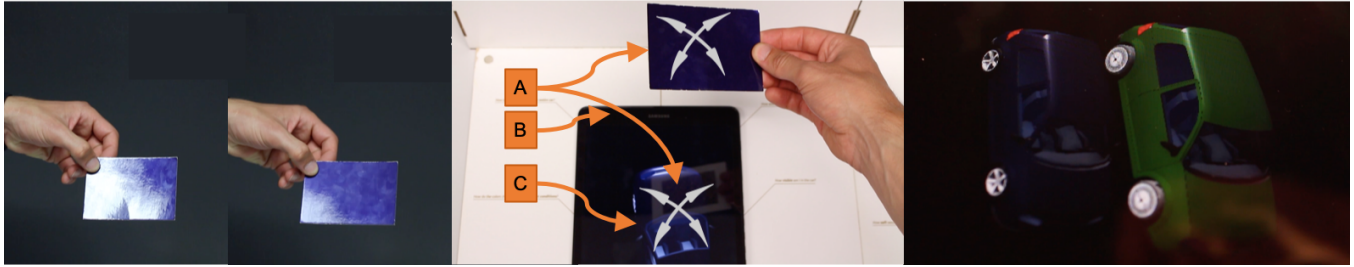


Figure 2: Exterior Paint Sample. Left: interaction with the material sample: tilting the sample to perceive the color in various light conditions. Middle: technical setup - (A) image marker on the back of the sample, (B) camera of the smart device, (C) digital 3D model of the car. Interaction- tilt the paint sample to tilt the digital car model. Right: comparing multiple cars on the smart device

and Klatzky [9], one moves fingers over the leather to feel, to perceive the texture of the material. That interaction with the sample to experience a particular aesthetic quality is the starting point of the connection with a digital layer. The context of our case is automotive retail. We present four regarding examples of our approach and design perspective. An important requirement of our work was to stay close to technology that customers commonly have at hand outside of the retail store, i.e., smart devices as tablets and mobile phones. In the following chapter we will highlight related work, and will then present the four material samples and designed interactions. This is followed by an impression what a visitor of our CHI demo can expect.

2 RELATED WORK

2.1 Extended Reality to Enhance the Retail Experience

The term *Phygital* is commonly used in retail to address hybrid shopping experiences: the physical and digital components at the same moment and in the same location [2]. Technological innovation has affected how products are offered and displayed, also outside of the physical stores [17]. Augmented Reality (AR) is found in applications such as Smart Mirrors, allowing customers to try on digital clothing [11], and applications for the home environment such as Ikea Place and Amazon AR View [10] that allow customers to place digital products in their physical home to get an impression of e.g. size and color fit to the exact imagined location.

2.2 Utilizing Mobile Phone Sensors

Researchers have explored a variety of technologies to extend interactions with mobile phones beyond the touch screen while using only passive add-ons. Sometimes, the mobile phone also uses its actuators (e.g. the vibration motor, or integrated LED's to enhance the sensing capabilities of the phone, see for example Spectrophone [13]). We specifically deem low-tech add-ons fitting to the setting of physical samples in the retail environment, as they should be low-cost to give away to customers, not requiring additional technology outside of the customer's mobile phone. Magnets are one way to interact with the mobile phone through static magnets and the magnetometer commonly integrated in mobile phones. Examples

of work that utilizes magnets to interact with the phone, include magnets in a pen [5], keyboard [1] or a phone case [4]). In these examples, the dynamics in magnetic strength and direction are used to capture movement. Sound can function as input to identify materials [14], but also to capture interaction such as pressure applied to a stylus when interacting with the smart phone [6], as well as buttons that can be used around the smart phone [8]. The camera sensor and activating LED's can be used to identify materials based on reflective qualities of the material surface [13], common used examples are the recognition of faces such as done by filters in smart phone applications such as Snapchat ¹, or surfaces to support AR [3], or in AR apps such as Ikea Place or Amazon AR View [10]. The accelerometer is used in combination with vibration activation to estimate the pressure input on mobile devices, capturing the vibration absorption [7]. Also as 'normal' input, e.g. to tilt the screen from portrait to landscape, or recognize the user faces the phone when looking at the screen. Capacitive touch on screen: capturing objects on the screen, e.g. PUCs [15], using capacitive material to mimic a touch, or using multiple to create a pattern to recognize. Also commonly used pens, or tips on gloves use this approach.

3 PHYGITAL SAMPLES

Phygital Samples digitally extend the experience that retail customers have when exploring specific physical qualities of a material sample. The interaction to explore that specific physical quality is concurrently the way of interacting with the digital extension aimed to provide a more complete experience of the whole product. In the following, we will describe four *phygital* samples we designed for the car retail context: one sample each for exterior paint, tinted windows, cushioning, and leather seat covering.

3.1 Exterior Paint Sample

Exterior paint of the car provides character to the car and is a definite consideration when purchasing a car. This *phygital* sample aims to assist in bridging the detailed view on the car paint on the sample to the global view of the color on the total car.

To experience the details of the material qualities of the color of the paint, the customer rotates the sample to see the color in different light conditions (see fig. 2, left). Particularly with metallic

¹Snapchat, Snap Inc., Santa Monica, California, U.S.A.

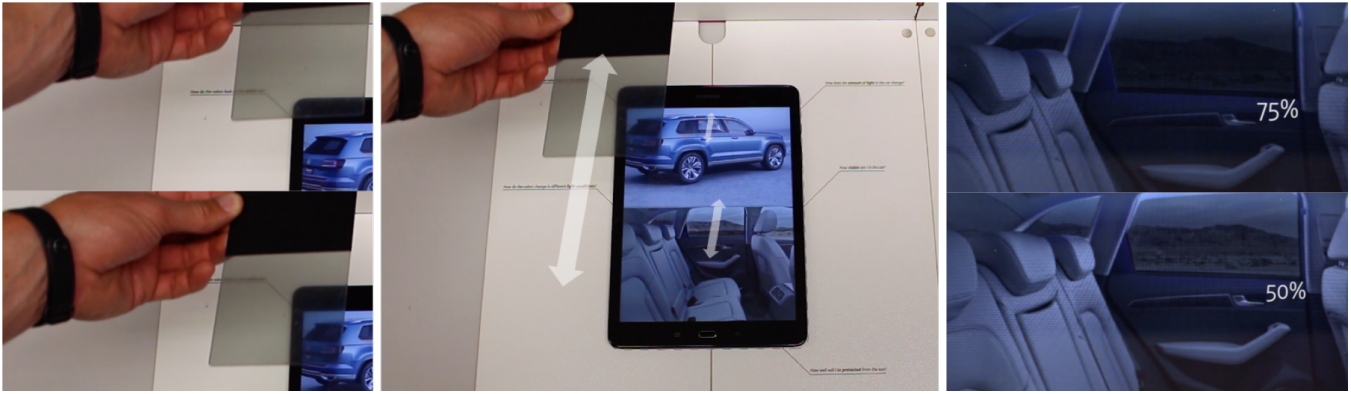


Figure 3: Tinted Glass Sample. Left: interaction with the material sample: alternately moving the sample in and outside of ones view, looking at an object without and through the tinted glass to see the difference in shade. Middle: interaction- hold the sample in the view of the camera to see tinted glass on an image of the exterior car and interior of the car. Move the sample vertically (in parallel to the smart device) to slide open the window of the car images. Right: multiple samples can be used to compare different shades of glass

samples one can experience much depth and subtle color differences. The middle image in figure 2 shows the phygital interaction. By holding the sample above the smart device, a digital 3D model of a car appears on the screen. Moving the sample to perceive the color in different light conditions concurrently rotates the virtual car. This creates a direct mapping, allowing the user to control and view the whole car from different angles and light conditions concurrently. Multiple samples can be used next to each other to compare different colors both physically and digitally (see fig. 2, right). The back of the sample carries a unique image marker (fig. 2, middle, (A)) that is recognized and tracked by the front camera (fig. 2, middle, (B)) of the smart device. The software implementation in Unity3D² captures the movements of the image marker, renders a 3D model of the car in the respective paint, (fig. 2, middle, (C)) and maps the movements of the sample to the digital model of the car on the screen.

3.2 Tinted Glass Sample

High-end car manufacturers offer various shades of tinted windows. These differ in opacity and darkness, ranging from lightly tinted to almost black. For buyers it might be difficult to envision the effect each tint has on the final looks of the car. Next to the exterior look of the car, it is important to consider how much of sunlight will get blocked by each shade, and how much one can see of the outside world when sitting in the car (and from outside in as well). On the screen two images are shown: the exterior of the car and the interior of the car. By moving the glass sample up and down, the windows of the car image slide up and down. This allows the user to see how the glass looks on the exterior of the car, as well as how it changes the light condition inside of the car and how the outside world looks like from within the car, while concurrently viewing the tint of the different physical sample. The implementation of this sample is similar to the Exterior paint sample; the back of the sample carries

a unique image target for tracking (1) that is read by the front camera (2) of the smart device. The software implementation in Unity3D captures the movements of the image target and renders the vertical movement of the digital car windows according to the vertical movement of the sample.

3.3 Seat Cushioning Sample

The haptic feeling of sitting on a seat of the car is an important aspect of the car experience. The outer layer of the seat significantly contributes to this haptic experience. This phygital sample aims to fill the gap between the haptic touch of a seat and a visual representation of sitting on the seat. As a full seat sample would be outside the scale of our envisioned (easy to take-away) samples, we chose for the haptics of pressing the cushioning with the hand. Pressing the top of the sample provides a tactile and haptic preview of the cushioning of the seat. On the screen the seat of the car is pressed in simulating a person sitting on the seat. We specifically chose for a 'sitting robot' to increase visibility of the press inside the seat material. This allows the customer to feel the seat cushioning, see the full seat in context, and see how it deforms when sat on. The cushioning sample is equipped with a static magnet, that moves down when the sample is pressed. The magnetometer of the phone is used to capture the movement of the magnet underneath the sample and this is mapped to video stills within the software. This creates a direct mapping between the press on the sample and the video showing the car seat pressed down.

3.4 Leather Sample

Car manufacturers offer a high variety of different types and colors of leather that cover the seats and or other elements of the car like the dashboard. Leather is a material that provides a high end feel, yet many contemporary buyers are interested in sustainability factors such as carbon footprint. This phygital sample aims to show the background of the particular leather and its production process. Move your fingers over the leather sample to feel the

²Unity3D, Unity Technologies, San Francisco, USA

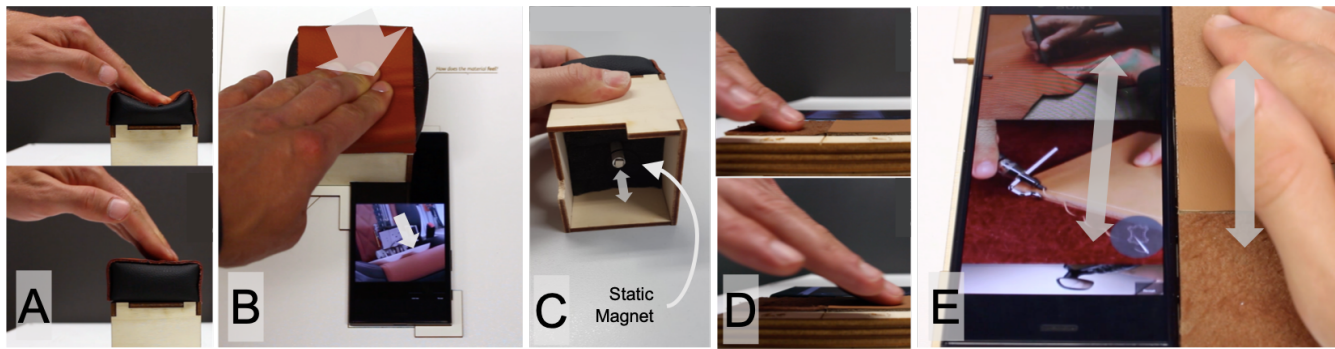


Figure 4: Seat Cushioning Sample (A,B,C) and Leather Sample (D,E). (A): interaction with the seat cushioning material sample. (B): press the cushion sample and the seat on the digital image will be pressed accordingly. (C): a build in magnet is pressed down and the change in magnetic field is sensed by the magnetometer. (D): interaction with the leather sample: lateral movement, i.e. moving the fingers over the material to perceive its texture. (E): the smart device shows content related to the physical sample currently touched

various tactile qualities that the leather has throughout the various stages of the production process. The digital layer shows images of the leather production process linked to each part of the production process. The leather sample is placed together with the phone on a seesaw mechanism that slightly tilts when the leather sample is touched at the bottom or top side. This slight tilting movement is barely noticeable by the user, but sensed by the smart device. The movement is detected by the accelerometer of the phone and the imagery slides to the according sample side - showing information that is relevant to the production stage sample that is touched at that moment.

4 DISCUSSION AND CONCLUSION

In summary, with this work we demonstrate a novel interaction technique to interact with material samples in the retail context. We call this approach *Phygital Samples*, aiming to integrate interacting with a physical material samples with a digital 'bigger picture'. As such, a connection is maintained between the sample and digital representation, creating a more integrated experience, in contrast to, for instance, combining material samples with an online car configurator. As in the latter the connection between both is missing and needs to be constituted by the customer. In *Phygital Samples*, the connection between the sample and the digital bigger picture is created through the interaction with the material sample. Following the Interaction Frogger Framework [16], physical input and digital output are mapped according to the six aspects of natural coupling. E.g., moving the tinted glass sample up, and the window of the digital car image will move up accordingly, in the same time, direction, expression, dynamics, and modality. The digital 'bigger picture' is selected based on contextual knowledge about what customers seek to experience in order to get a good understanding of a products material qualities, e.g., a material sample of exterior paint provides a detailed, accurate representation of actual material qualities, while the digital augmentation allows to position the choice in a bigger picture. Through maintaining a meaningful and constant coupling between the physical sample and the digital layer, our *Phygital Samples* approach aims to create rich material experiences that closely resemble actual product configurations.

REFERENCES

- [1] Heba Abdelnasser, Khaled A. Harras, and Moustafa Youssef. 2020. MagStroke: A Magnetic Based Virtual Keyboard for Off-the-Shelf Smart Devices. In *2020 17th Annual IEEE International Conference on Sensing, Communication, and Networking (SECON)*. 1–9. <https://doi.org/10.1109/SECON48991.2020.9158416>
- [2] Sara Belghiti, Adeline Ochs, Jean-François Lemoine, and Olivier Badot. 2018. The phygital shopping experience: an attempt at conceptualization and empirical investigation. In *Academy of marketing science world marketing congress*. Springer, 61–74.
- [3] Anders Henrysson, Mark Ollila, and Mark Billinghurst. 2005. Mobile Phone Based AR Scene Assembly. In *Proceedings of the 4th International Conference on Mobile and Ubiquitous Multimedia* (Christchurch, New Zealand) (MUM '05). Association for Computing Machinery, New York, NY, USA, 95–102. <https://doi.org/10.1145/1149488.1149504>
- [4] Sungjae Hwang, Myungwook Ahn, and Kwangyun Wahn. 2013. Magnetic Marionette: Magnetically Driven Elastic Controller on Mobile Device. In *Proceedings of the Companion Publication of the 2013 International Conference on Intelligent User Interfaces Companion* (Santa Monica, California, USA) (IUI '13 Companion). Association for Computing Machinery, New York, NY, USA, 75–76. <https://doi.org/10.1145/2451176.2451207>
- [5] Sungjae Hwang, Andrea Bianchi, Myungwook Ahn, and Kwangyun Wahn. 2013. MagPen: Magnetically Driven Pen Interactions on and around Conventional Smartphones (*MobileHCI '13*). Association for Computing Machinery, New York, NY, USA, 412–415. <https://doi.org/10.1145/2493190.2493194>
- [6] Sungjae Hwang, Andrea Bianchi, and Kwangyun Wahn. 2012. MicPen: Pressure-Sensitive Pen Interaction Using Microphone with Standard Touchscreen. In *CHI '12 Extended Abstracts on Human Factors in Computing Systems* (Austin, Texas, USA) (CHI EA '12). Association for Computing Machinery, New York, NY, USA, 1847–1852. <https://doi.org/10.1145/2212776.2223717>
- [7] Sungjae Hwang, Andrea Bianchi, and Kwang-yun Wahn. 2013. VibPress: Estimating Pressure Input Using Vibration Absorption on Mobile Devices. In *Proceedings of the 15th International Conference on Human-Computer Interaction with Mobile Devices and Services* (Munich, Germany) (*MobileHCI '13*). Association for Computing Machinery, New York, NY, USA, 31–34. <https://doi.org/10.1145/2493190.2493193>
- [8] Sungjae Hwang and Kwang-yun Wahn. 2012. PseudoButton: Enabling Pressure-Sensitive Interaction by Repurposing Microphone on Mobile Device. In *CHI '12 Extended Abstracts on Human Factors in Computing Systems* (Austin, Texas, USA) (CHI EA '12). Association for Computing Machinery, New York, NY, USA, 1565–1570. <https://doi.org/10.1145/2212776.2223673>
- [9] Susan J Lederman and Roberta L Klatzky. 2009. Haptic perception: A tutorial. *Attention, Perception, & Psychophysics* 71, 7 (2009), 1439–1459.
- [10] Jun-hyuk Lee and Seung-In Kim. 2019. Evaluation of user experience in AR-based shopping applications-Focused on Ikea place and Amazon AR View. *Journal of Digital Convergence* 17, 10 (2019), 411–416.
- [11] Ayotunde Ogunjimi, Mizan Rahman, Nazrul Islam, and Rajibul Hasan. 2021. Smart mirror fashion technology for the retail chain transformation. *Technological Forecasting and Social Change* 173 (2021), 121118.

- [12] Eleonora Pantano. 2014. Innovation drivers in retail industry. *International Journal of Information Management* 34, 3 (2014), 344–350.
- [13] Maximilian Schrapel, Philipp Etgeton, and Michael Rohs. 2021. SpectroPhone: Enabling Material Surface Sensing with Rear Camera and Flashlight LEDs. In *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) (CHI EA '21). Association for Computing Machinery, New York, NY, USA, Article 336, 5 pages. <https://doi.org/10.1145/3411763.3451753>
- [14] Seok-Jeong Song and Hyoungsik Nam. 2018. Sound-of-Tapping user interface technology with medium identification. *Displays* 53 (2018), 54–64. <https://doi.org/10.1016/j.displa.2018.02.003>
- [15] Simon Voelker, Kosuke Nakajima, Christian Thoresen, Yuichi Itoh, Kjell Ivar Øvergård, and Jan Borchers. 2013. PUCs: Detecting Transparent, Passive Untouched Capacitive Widgets on Unmodified Multi-Touch Displays. In *Proceedings of the 2013 ACM International Conference on Interactive Tabletops and Surfaces* (St. Andrews, Scotland, United Kingdom) (ITS '13). Association for Computing Machinery, New York, NY, USA, 101–104. <https://doi.org/10.1145/2512349.2512791>
- [16] S. A. G. Wensveen, J. P. Djajadiningrat, and C. J. Overbeeke. 2004. Interaction Frogger: A Design Framework to Couple Action and Function through Feedback and Feedforward. In *Proceedings of the 5th Conference on Designing Interactive Systems: Processes, Practices, Methods, and Techniques* (Cambridge, MA, USA) (DIS '04). Association for Computing Machinery, New York, NY, USA, 177–184. <https://doi.org/10.1145/1013115.1013140>
- [17] Kim Willems, Annelien Smolders, Malaika Brengman, Kris Luyten, and Johannes Schöning. 2017. The path-to-purchase is paved with digital opportunities: An inventory of shopper-oriented retail technologies. *Technological Forecasting and Social Change* 124 (2017), 228–242.

Received 19 January 2023